

Literature Review

Chess has long served as a useful testbed for studying artificial intelligence because it combines clear rules and objective outcomes with a large range of possible positions and levels of difficulty. With the rise of large language models (LLMs), chess has again become useful for evaluating reasoning because general models must understand board states, recognize tactical patterns, calculate sequences of moves, and make strategic decisions rather than rely on a specific search and prediction algorithm. Recent benchmarks, however, show that even strong LLMs continue to struggle with these tasks (Kolasani et al., 2025; Liu et al., 2026; Wen et al., 2025). This paper reviews research on human chess puzzle difficulty, LLM chess reasoning, and the characteristics of chess positions that may cause human and LLM performance to differ.

Chess puzzles provide a useful setting for comparing these two forms of difficulty because platforms such as Lichess and ChessTempo collect large amounts of solving data from human players (Ding et al., 2024; Hristova et al., 2014). These data allow researchers to estimate how difficult individual puzzles are for humans rather than relying only on expert judgment. At the same time, recent studies have evaluated LLMs on many of the same types of chess positions. Together, these areas of research raise an important question: whether puzzles that are difficult for humans are also difficult for LLMs, and what characteristics of a position may explain cases where the two differ.

Before comparing human and LLM difficulty, it is important to understand what makes a chess puzzle difficult for a human in the first place. Hristova et al. (2014) studied tactical chess problems using ChessTempo ratings, solving accuracy, solving time, expert judgments, and information about how players examined the positions. One important result was that even experienced chess players were not especially accurate at predicting which puzzles would be difficult for other players. This supports the use

of empirical ratings based on the performance of many solvers rather than relying only on a small number of expert opinions.

The study also showed that puzzle difficulty cannot be understood simply by looking at how complicated a board appears. Solvers must recognize relevant tactical ideas and then calculate whether those ideas actually work. For the players studied by Hristova et al. (2014), correctly calculating variations played an important role in whether a puzzle was successfully solved. A position with relatively few pieces may therefore still be difficult if the correct continuation is unexpected or if several reasonable alternatives must be considered and rejected. This suggests that the difficulty of a chess puzzle depends not only on visible board characteristics but also on the reasoning required to reach the solution.

Stoiljkovikj et al. (2015) extended this idea by asking whether human puzzle difficulty could be estimated computationally. Their approach used a search tree designed to represent the moves and variations that a human solver would realistically consider rather than every legal continuation available in the position. Characteristics of these search trees were then used to classify chess problems by difficulty. Their results suggest that the structure of the solving process itself contains useful information about human difficulty. This work shifts attention away from simple measures of board complexity toward the choices a solver must make.

More recent research has begun to connect human puzzle difficulty with LLM performance. Ding et al. (2024) examined difficulty through the E2H-Lichess dataset, where puzzle difficulty is estimated using human solving behavior and Glicko-2 ratings. Across most models, accuracy decreases as puzzle difficulty increases, suggesting that human difficulty contains meaningful information about the challenges faced by LLMs. At the same time, the relationship is not uniform. GPT-4-Turbo performs particularly well on easier puzzles, while Claude 3 Opus performs comparatively better on difficult puzzles, showing that models can respond differently to the same human difficulty scale. Therefore, human-rated difficulty may be an important predictor of LLM performance without fully determining it.

Rather than asking which move is objectively strongest, McIlroy-Young et al. (2020) looked at whether chess engines could predict the moves humans actually make. Traditional engines such as Stockfish and Leela did not closely reproduce human decisions at particular skill levels, even when their playing strength was reduced. In contrast, Maia, which was trained directly on human games, achieved higher move accuracy and could be targeted toward specific human rating ranges. This distinction is important because it shows that strong chess performance and human-like chess behavior are separate goals. In the context of puzzle difficulty, an LLM may therefore find a position difficult for reasons that differ from those that make the same position difficult for a human player.

Together, these studies suggest that human and machine difficulty are related but should not be assumed to be identical. Human difficulty provides a useful starting point for predicting LLM performance, but differences in how humans and artificial systems process chess positions may lead to important disagreements at each individual puzzle level.

Recent LLM chess benchmarks suggest several reasons why these disagreements may occur. ChessQA separates chess understanding into structural knowledge, tactical motifs, short tactical calculation, positional judgment, and semantic understanding. Its error analysis finds recurring problems such as misreading board states, producing illegal tactical moves, arriving at the wrong final move despite plausible analysis, and incorrectly determining that no valid answer exists (Wen et al., 2025). ChessArena similarly finds weaknesses in instruction following, tactical reasoning, and maintaining coherent reasoning across multiple turns. Even models with strong general reasoning abilities remain below Maia-1100 in full chess gameplay (Liu et al., 2026). LLM CHESS also shows that performance can change substantially depending on whether models receive legal moves, how the board is represented, and how the interaction is structured (Kolasani et al., 2025).

These findings suggest that LLMs can fail chess positions for reasons that are not identical to the difficulties faced by human players. Research on human puzzle difficulty places strong emphasis on

recognizing patterns and calculating meaningful alternatives. LLM research identifies additional problems involving board representation, legality, state tracking, and converting an apparently reasonable analysis into the correct final move. Therefore, an LLM's failure on a puzzle does not necessarily mean that the underlying tactical concept is difficult in the same way it is for a human.

The type of chess reasoning required may also matter. Wang et al. (2025) divide expert chess reasoning into long-term strategic factors, including material, piece activity, pawn structure, space, and king safety, and short-term tactical calculation. Using the MATE dataset, they find that providing language explanations of these strategic and tactical concepts improves LLM move selection, with the combination of strategy and tactics generally producing the strongest results. ChessQA similarly organizes Lichess tactical puzzles by both human puzzle rating and themes such as forks, pins, sacrifices, discovered attacks, promotions, and mating patterns (Wen et al., 2025). Together, these studies suggest that puzzles with similar human difficulty ratings may still place different demands on an LLM depending on their tactical theme, board structure, solution length, or type of reasoning required.

More recent work provides direct evidence that characteristics of chess positions can be used to predict human puzzle difficulty. Schütt et al. (2024) use a large collection of Lichess puzzles to estimate puzzle difficulty without relying only on the past solving history of each individual puzzle. Their neural network model predicts difficulty while also learning to identify the correct move and the puzzle's tactical themes. The authors additionally use uncertainty about the correct move as information for estimating difficulty. In other words, if the correct solution is difficult to distinguish from other reasonable moves, that uncertainty can itself indicate a harder puzzle.

This result connects closely with earlier work on meaningful search. Both approaches suggest that the existence of plausible alternatives may be an important source of difficulty. However, Schütt et al. (2024) approach the problem using modern machine learning and a much larger set of Lichess positions.

Their work shows that tactical themes and information about the solution provide additional predictive value beyond a simple difficulty label.

Björkqvist (2025) further expands this idea by combining human-like chess modeling, chess engines, and hand-crafted characteristics of Lichess puzzles. The study considers features including solution length, tactical themes, number of legal moves, checks, captures, material advantage, mobility, undefended pieces, pawn structure, king safety, sacrifices, recaptures, and mate threats. The final model combines these position features with information from systems such as Maia, Stockfish, and Leela. The results show that both human-like model behavior and specific characteristics of the puzzle provide useful information for predicting human difficulty.

These studies are especially relevant because they show that many possible explanations for difficulty can be measured directly. Earlier work focused on the structure of the search required to solve a puzzle, while more recent research has added tactical themes, move uncertainty, material and positional characteristics, and human-like model predictions. However, the main goal of these studies is still to predict human puzzle difficulty. They do not focus on whether those same characteristics explain situations where LLM difficulty differs from human difficulty. Research on chess puzzle difficulty generally attempts to explain or predict human difficulty, while research on LLM chess mainly evaluates overall model performance or identifies categories of LLM errors. Less is known about which characteristics of a chess position explain cases where an LLM performs substantially better or worse than would be expected from the puzzle's human difficulty rating.

This project addresses that gap by first examining the relationship between human-rated puzzle difficulty and LLM success and then investigating whether characteristics such as tactical theme, solution length, number of legal moves, game phase, number of pieces, and other features of the position help explain disagreements between human and LLM difficulty.

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